Standard Cell Library Design and Characterization

SKY130 PDK — Project 1

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Abstract

This report documents the design and SPICE-based characterization of a 13-cell standard cell library using the SkyWater SKY130 open-source PDK. The library includes inverters, NAND2, NOR2, and MAJ3 gates at multiple drive strengths. Non-Linear Delay Model (NLDM) tables are extracted under typical-typical (TT) corner conditions (V_{DD} = 1.8 V, T = 25 °C) using an automated Python/ngspice characterization flow. Simulation results are compared against a first-order RC delay model to assess its accuracy and limitations.

GitHub Repository: https://github.com/Sama3mad/Project-1---Standard-Cell-Library-Design-and-Characterization-.git

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1 Technology Parameters from Tutorial 2

This section consolidates all SKY130 technology parameters extracted in Tutorial 2. These values are used directly in Phase 1 cell sizing and in the analytical RC model comparison (Section 6).

1.1 SKY130 Minimum Device Dimensions

The SKY130 process minimum feature sizes are:

Parameter	Symbol	Value
Minimum channel length	L_{min}	0.15 µm
Minimum NMOS width	$W_{N,\text{min}}$	0.42 µm
Matched PMOS width	$W_{P,\text{min}}$	1.00 µm

1.2 Effective Resistance Ratio K_p

K_p is the ratio of the effective channel resistance of the smallest PMOS to that of the smallest NMOS, measured by timing the charge/discharge of a load capacitor:

$$K_p = \frac{R_{P,\text{eff}}}{R_{N,\text{eff}}} = \frac{T_P}{T_N}$$

Simulated K_p Result (Tutorial 2, Section 2.1)

$$T_N = 1.240 \times 10^{-10}\,\text{s} \quad \text{(NMOS discharges } C_L = 25\,\text{fF})$$
$$T_P = 2.943 \times 10^{-10}\,\text{s} \quad \text{(PMOS charges } C_L = 25\,\text{fF})$$

$$\boxed{K_p = 2.373}$$

To achieve a *symmetrical* inverter (equal rise/fall times), the PMOS width must be scaled up by K_p relative to the NMOS:

$$W_P = K_p \times W_N = 2.373 \times 0.42\,\mu\text{m} \approx 1.00\,\mu\text{m}$$

1.3 Effective Transistor Resistances

Using $R = T/(0.69 \times C_L)$ with $C_L = 25\,\text{fF}$:

Transistor	Formula	Value
Smallest NMOS ($W = 0.42\,\mu\text{m}$)	$R_N = T_N/(0.69 \times C_L)$	$\approx 7.19\,\text{k}\Omega$
Matched PMOS ($W = 1.00\,\mu\text{m}$)	$R_P = T_P/(0.69 \times C_L)$	$\approx 17.07\,\text{k}\Omega$

Note on R_N

$R_N = T_N/(0.69\,C_L) = 1.240 \times 10^{-10}/(0.69 \times 25 \times 10^{-15}) \approx 7.19\,\text{k}\Omega$. This is the effective switch resistance of the minimum-sized NMOS; it is *not* the MOS transconductance parameter μC_{ox}.

1.4 Smallest Inverter Propagation Delay t_{inv}

The propagation delay of the smallest symmetrical inverter driving an identical inverter (loaded by C_{inv}) was measured in Tutorial 2, Section 2.2:

Simulated t_{inv} (Tutorial 2, Section 2.2)

$$t_{pdr} = 19.79\,\text{ps} \quad \text{(output rising)}$$
$$t_{pdf} = 22.34\,\text{ps} \quad \text{(output falling)}$$
$$\boxed{t_{p,\text{avg}} = 21.07\,\text{ps}}$$

1.5 Inverter Input Capacitance C_{inv}

Measured via the RC time-constant method (Tutorial 2, Section 2.3):

$$C_{inv} = \frac{\Delta t_{RC}}{0.69 \times R_{drv}} = \frac{11.02\,\text{ps}}{0.69 \times 10\,\text{k}\Omega}$$

C_{inv} Result

$$\boxed{C_{inv} \approx 1.60\,\text{fF}}$$

This value sets the base load for one inverter input and is used when computing FO4 delay and RC model predictions.

1.6 Summary of Technology Parameters

Parameter	Symbol	Value	Source
Min. channel length	L_{min}	0.15 µm	SKY130 DRC
Min. NMOS width	$W_{N,\text{min}}$	0.42 µm	SKY130 DRC
Matched PMOS width	$W_{P,\text{min}}$	1.00 µm	Tut. 2 - 2.2
Effective resistance ratio	K_p	2.373	Tut. 2 - 2.1
Min. NMOS eff. resistance	R_N	7.19 kΩ	Tut. 2 - 2.1
Min. PMOS eff. resistance	R_P	17.07 kΩ	Tut. 2 - 2.1
Inverter input capacitance	C_{inv}	1.60 fF	Tut. 2 - 2.3
Smallest inverter avg. delay	$t_{p,\text{min}}$	21.07 ps	Tut. 2 - 2.2
Supply voltage	V_{DD}	1.8 V	SKY130 spec
Temperature	T	25 °C	TT corner

2 Phase 1 — Cell Sizing

2.1 Sizing Methodology

Every cell in the library is sized so that its output rises and falls at roughly the same speed — this is called **symmetric** timing. We do this for the base (x1) version of each cell, then simply multiply the transistor widths to get the stronger drive versions (x2, x4, x8).

Three rules cover all 13 cells:

1. **Base NMOS width** is always $W_{N,\min} = 0.42\,\mu\text{m}$. If k NMOS transistors are connected in series (as in NAND gates), each one is made k times wider so the combined resistance stays the same as a single transistor.

2. **Base PMOS width** is $K_p \times W_{N,\min} \approx 1.00\,\mu\text{m}$, because PMOS is naturally $K_p = 2.373$ times slower than NMOS at the same size. If k PMOS transistors are in series (as in NOR gates), the same k-times widening is applied.

3. **Drive-strength scaling:** to get x2, x4, or x8, multiply *both* W_N and W_P by that factor. The ratio W_P/W_N never changes, so symmetry is preserved.

Channel length is fixed at $L = 0.15\,\mu\text{m}$ for every transistor.

2.2 Inverter Family (INV)

$$Y = \overline{A}$$

An inverter has one PMOS on top (pull-up) and one NMOS on the bottom (pull-down). There are no series stacks, so no extra widening is needed beyond the K_p ratio.

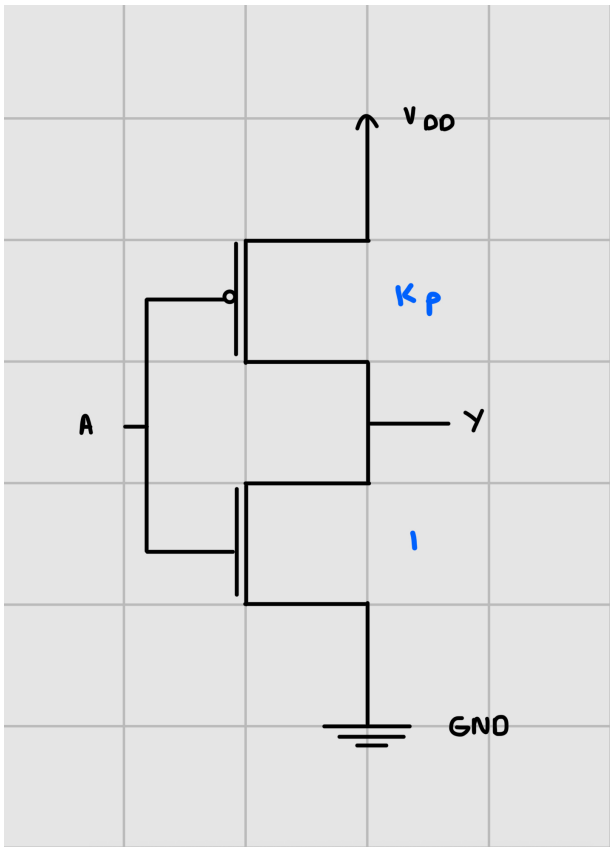

Figure 1: Inverter (INV) schematic. PMOS (top) pulls the output high; NMOS (bottom) pulls it low. Input A drives both gates.

Cell	W_N (µm)	W_P (µm)	L (µm)	Drive strength
invx1	0.42	1.00	0.15	×1
invx2	0.84	2.00	0.15	×2
invx4	1.68	4.00	0.15	×4
invx8	3.36	8.00	0.15	×8

Sizing rationale: No series stacks, so $W_N = k \times 0.42\,\mu\text{m}$ and $W_P = k \times 1.00\,\mu\text{m}$ for drive-strength multiplier k.

2.3 NAND2 Family

$$Y = \overline{A \cdot B}$$

A NAND2 gate pulls the output *low* only when *both* A and B are high. This requires the two NMOS transistors to be in **series** — both must be ON at the same time. Two transistors in series double the pull-down resistance, so each NMOS is made 2× wider. The two PMOS are in **parallel** so either one can pull the output high — no extra widening needed.

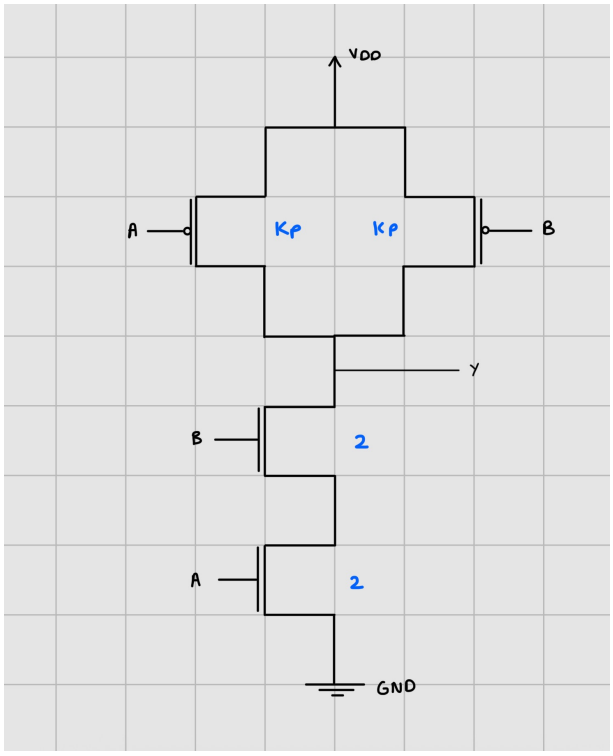

Figure 2: NAND2 schematic. Two PMOS in parallel (pull-up); two NMOS in series (pull-down).

$$W_{N,\mathrm{x1}} = 2 \times 0.42\,\mu\mathrm{m} = 0.84\,\mu\mathrm{m} \qquad W_{P,\mathrm{x1}} = 1 \times 1.00\,\mu\mathrm{m} = 1.00\,\mu\mathrm{m}$$

Cell	W_N (µm)	W_P (µm)	L (µm)	Drive strength
nand2x1	0.84	1.00	0.15	×1
nand2x2	1.68	2.00	0.15	×2
nand2x4	3.36	4.00	0.15	×4

2.4 NOR2 Family

$$Y = \overline{A + B}$$

A NOR2 gate pulls the output *high* only when *both* A and B are low. This requires the two PMOS transistors to be in **series**. Two PMOS in series double the pull-up resistance, so each is made 2× wider. The two NMOS are in **parallel** so either can pull the output low — no extra widening needed.

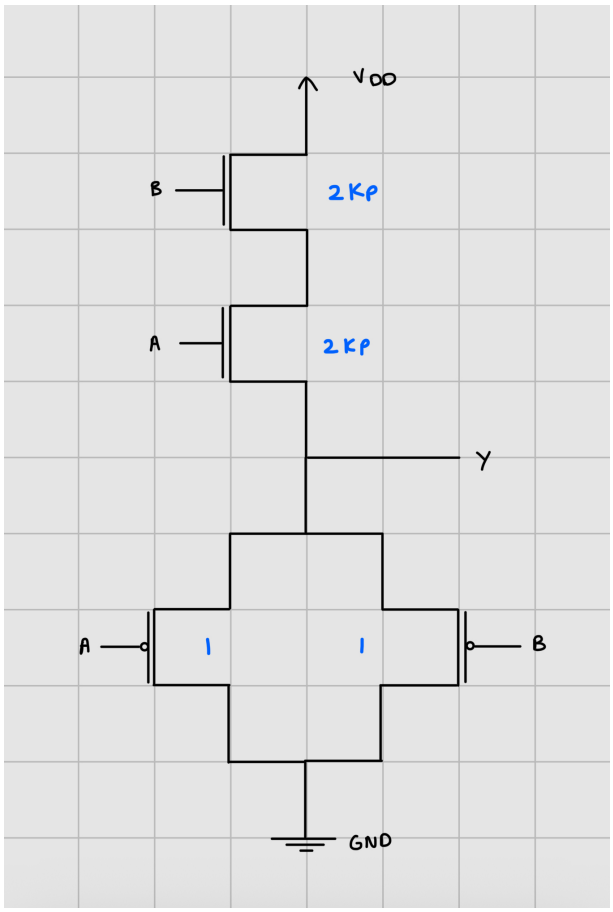

Figure 3: NOR2 schematic. Two PMOS in series (pull-up); two NMOS in parallel (pull-down).

$$W_{P,\mathrm{x1}} = 2 \times 1.00\,\mu\mathrm{m} = 2.00\,\mu\mathrm{m} \qquad W_{N,\mathrm{x1}} = 1 \times 0.42\,\mu\mathrm{m} = 0.42\,\mu\mathrm{m}$$

Cell	W_N (µm)	W_P (µm)	L (µm)	Drive strength
nor2x1	0.42	2.00	0.15	×1
nor2x2	0.84	4.00	0.15	×2
nor2x4	1.68	8.00	0.15	×4

2.5 MAJ3 Family

$$Y = \overline{AB + BC + AC}$$

The majority gate outputs 1 when at least two of the three inputs (A, B, C) are 1.

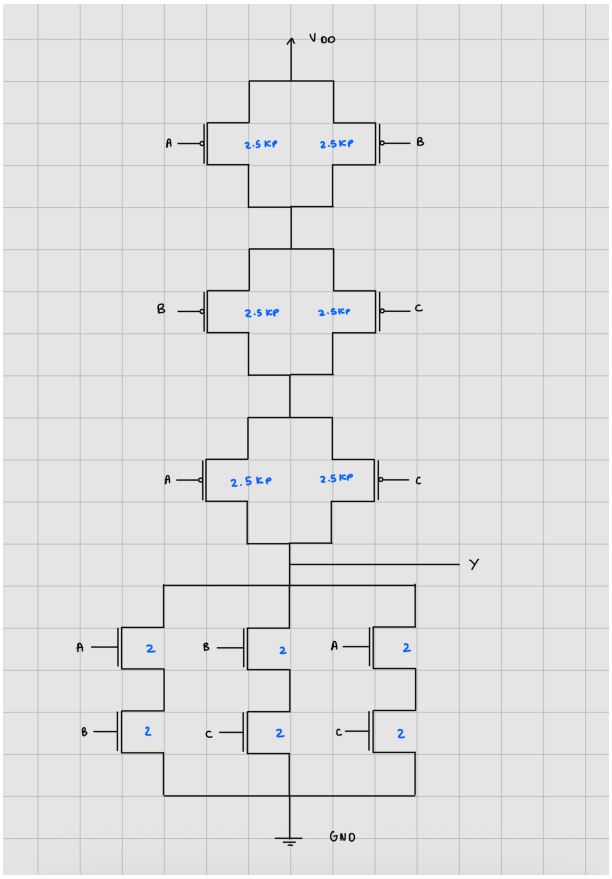

Figure 4: MAJ3 (AOI) schematic. Three PMOS in parallel form the pull-up; three pairs of series NMOS (branches AB, BC, AC) form the pull-down.

Sizing rationale

Pull-down (NMOS). Each of the three branches contains 2 NMOS in series. A depth-2 stack doubles the resistance, so each NMOS must be 2× wider:

$$W_{N,\mathrm{x1}} = 2 \times 0.42\,\mu\mathrm{m} = 0.84\,\mu\mathrm{m}$$

Pull-up (PMOS). The pull-up network consists of three series stages, each containing two PMOS in parallel. The effective resistance depends on how many parallel devices conduct in each stage.

To achieve a symmetric design, we match the worst-case pull-up resistance to that of the pull-down network. Using resistance equivalence:

$$\frac{1}{xR} + \frac{1}{xR} + \frac{1}{2xR} = \frac{1}{R}$$

Solving gives:

$$x = 2.5$$

Thus, the PMOS width is scaled as:

$$W_{P,\mathrm{x1}} = 2.5 \times K_p \times W_{N,\min} \approx 2.50\,\mu\mathrm{m}$$

Cell	W_N (µm)	W_P (µm)	L (µm)	Drive strength
maj3x1	0.84	2.50	0.15	×1
maj3x2	1.68	5.00	0.15	×2
maj3x4	3.36	10.00	0.15	×4

2.6 Complete Sizing Summary

Cell	Function	W_N (µm)	W_P (µm)	L (µm)	Series stack
invx1	$\bar{A}$	0.42	1.00	0.15	none
invx2	$\bar{A}$	0.84	2.00	0.15	none
invx4	$\bar{A}$	1.68	4.00	0.15	none
invx8	$\bar{A}$	3.36	8.00	0.15	none
nand2x1	$\overline{AB}$	0.84	1.00	0.15	2 NMOS
nand2x2	$\overline{AB}$	1.68	2.00	0.15	2 NMOS
nand2x4	$\overline{AB}$	3.36	4.00	0.15	2 NMOS
nor2x1	$\overline{A+B}$	0.42	2.00	0.15	2 PMOS
nor2x2	$\overline{A+B}$	0.84	4.00	0.15	2 PMOS
nor2x4	$\overline{A+B}$	1.68	8.00	0.15	2 PMOS
maj3x1	$\overline{AB+BC+AC}$	0.84	2.50	0.15	2 NMOS, 2.5× PMOS
maj3x2	$\overline{AB+BC+AC}$	1.68	5.00	0.15	2 NMOS, 2.5× PMOS
maj3x4	$\overline{AB+BC+AC}$	3.36	10.00	0.15	2 NMOS, 2.5× PMOS

3 Phase 2 — SPICE Library

In this phase, we wrote the SPICE subcircuit library containing all 13 standard cells and verified that each cell produces the correct logic output before moving on to characterization.

3.1 Library File Structure

The library is split into two files to avoid nested-`.lib` errors in ngspice:

- `cells.lib` — contains only `.subckt` definitions, no model includes.
- `verify.sp` (and later `characterize.py`) — the top-level deck that loads the SKY130 models and then includes `cells.lib`.

Every top-level deck begins with the following three lines before anything else:

```
.lib /home/sama/work/pdks/volare/sky130/versions/
     dd7771c384ed36b91a25e9f8b314355fc26561be/
     sky130A/libs.tech/ngspice/sky130.lib.spice tt
.temp 25
.include cells.lib
```

Listing 1: Required header in every top-level deck

The `tt` corner selects the typical-typical device models. `.temp 25` fixes the simulation temperature to 25 °C, consistent with the TT characterization corner defined for this project.

3.2 Subcircuit Definitions

All 13 cells share the same conventions:

- Channel length fixed at $L = 0.15\,\mu\text{m}$ for every transistor.
- 1.8 V core device models: `sky130_fd_pr__nfet_01v8` (NMOS) and `sky130_fd_pr__pfet_01v8` (PMOS).
- Pin order: `inputs... Y VDD VSS`.
- Widths follow the sizing derived in Section 2.

```
* ============================================================
* Standard Cell Library     SKY130 TT Corner
* L = 0.15 um for all transistors
* Pin order: inputs... Y VDD VSS
* ============================================================

*        Inverter Family



.subckt invx1 A Y VDD VSS
XMN Y A VSS VSS sky130_fd_pr__nfet_01v8 w=0.42 l=0.15
XMP Y A VDD VDD sky130_fd_pr__pfet_01v8 w=1.00 l=0.15
.ends invx1

.subckt invx2 A Y VDD VSS
XMN Y A VSS VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMP Y A VDD VDD sky130_fd_pr__pfet_01v8 w=2.00 l=0.15
.ends invx2

.subckt invx4 A Y VDD VSS
XMN Y A VSS VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMP Y A VDD VDD sky130_fd_pr__pfet_01v8 w=4.00 l=0.15
.ends invx4

.subckt invx8 A Y VDD VSS
XMN Y A VSS VSS sky130_fd_pr__nfet_01v8 w=3.36 l=0.15
```

```
XMP Y A VDD VDD sky130_fd_pr__pfet_01v8 w=8.00 l=0.15
.ends invx8

*        NAND2 Family


* Pull-down: 2 NMOS in series  (A top, B bottom)
* Pull-up  : 2 PMOS in parallel

.subckt nand2x1 A B Y VDD VSS
XMN1 Y   A mid VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMN2 mid B VSS VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMP1 Y   A VDD VDD sky130_fd_pr__pfet_01v8 w=1.00 l=0.15
XMP2 Y   B VDD VDD sky130_fd_pr__pfet_01v8 w=1.00 l=0.15
.ends nand2x1

.subckt nand2x2 A B Y VDD VSS
XMN1 Y   A mid VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMN2 mid B VSS VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMP1 Y   A VDD VDD sky130_fd_pr__pfet_01v8 w=2.00 l=0.15
XMP2 Y   B VDD VDD sky130_fd_pr__pfet_01v8 w=2.00 l=0.15
.ends nand2x2

.subckt nand2x4 A B Y VDD VSS
XMN1 Y   A mid VSS sky130_fd_pr__nfet_01v8 w=3.36 l=0.15
XMN2 mid B VSS VSS sky130_fd_pr__nfet_01v8 w=3.36 l=0.15
XMP1 Y   A VDD VDD sky130_fd_pr__pfet_01v8 w=4.00 l=0.15
XMP2 Y   B VDD VDD sky130_fd_pr__pfet_01v8 w=4.00 l=0.15
.ends nand2x4

*        NOR2 Family


* Pull-up  : 2 PMOS in series
* Pull-down: 2 NMOS in parallel

.subckt nor2x1 A B Y VDD VSS
XMP1 mid A VDD VDD sky130_fd_pr__pfet_01v8 w=2.00 l=0.15
XMP2 Y   B mid mid sky130_fd_pr__pfet_01v8 w=2.00 l=0.15
XMN1 Y   A VSS VSS sky130_fd_pr__nfet_01v8 w=0.42 l=0.15
XMN2 Y   B VSS VSS sky130_fd_pr__nfet_01v8 w=0.42 l=0.15
.ends nor2x1

.subckt nor2x2 A B Y VDD VSS
XMP1 mid A VDD VDD sky130_fd_pr__pfet_01v8 w=4.00 l=0.15
XMP2 Y   B mid mid sky130_fd_pr__pfet_01v8 w=4.00 l=0.15
XMN1 Y   A VSS VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMN2 Y   B VSS VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
.ends nor2x2

.subckt nor2x4 A B Y VDD VSS
```

```
XMP1 mid A VDD VDD sky130_fd_pr__pfet_01v8 w=8.00 l=0.15
XMP2 Y   B mid mid sky130_fd_pr__pfet_01v8 w=8.00 l=0.15
XMN1 Y   A VSS VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMN2 Y   B VSS VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
.ends nor2x4

*        MAJ3 Family  (AOI: Y = NOT(AB + BC + AC))

* Pull-down: 3 parallel branches, each with 2 NMOS in series
* Pull-up  : shared node mPA with PMOS parallel+series network

.subckt maj3x1 A B C Y VDD VSS
XMN_AB1 Y   A mAB VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMN_AB2 mAB B VSS VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMN_BC1 Y   B mBC VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMN_BC2 mBC C VSS VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMN_AC1 Y   A mAC VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMN_AC2 mAC C VSS VSS sky130_fd_pr__nfet_01v8 w=0.84 l=0.15
XMP_A1  mPA A VDD VDD sky130_fd_pr__pfet_01v8 w=2.50 l=0.15
XMP_B1  mPA B VDD VDD sky130_fd_pr__pfet_01v8 w=2.50 l=0.15
XMP_A2  Y   A mPA mPA sky130_fd_pr__pfet_01v8 w=2.50 l=0.15
XMP_C1  Y   C mPA mPA sky130_fd_pr__pfet_01v8 w=2.50 l=0.15
.ends maj3x1

.subckt maj3x2 A B C Y VDD VSS
XMN_AB1 Y   A mAB VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMN_AB2 mAB B VSS VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMN_BC1 Y   B mBC VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMN_BC2 mBC C VSS VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMN_AC1 Y   A mAC VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMN_AC2 mAC C VSS VSS sky130_fd_pr__nfet_01v8 w=1.68 l=0.15
XMP_A1  mPA A VDD VDD sky130_fd_pr__pfet_01v8 w=5.00 l=0.15
XMP_B1  mPA B VDD VDD sky130_fd_pr__pfet_01v8 w=5.00 l=0.15
XMP_A2  Y   A mPA mPA sky130_fd_pr__pfet_01v8 w=5.00 l=0.15
XMP_C1  Y   C mPA mPA sky130_fd_pr__pfet_01v8 w=5.00 l=0.15
.ends maj3x2

.subckt maj3x4 A B C Y VDD VSS
XMN_AB1 Y   A mAB VSS sky130_fd_pr__nfet_01v8 w=3.36 l=0.15
XMN_AB2 mAB B VSS VSS sky130_fd_pr__nfet_01v8 w=3.36 l=0.15
XMN_BC1 Y   B mBC VSS sky130_fd_pr__nfet_01v8 w=3.36 l=0.15
XMN_BC2 mBC C VSS VSS sky130_fd_pr__nfet_01v8 w=3.36 l=0.15
XMN_AC1 Y   A mAC VSS sky130_fd_pr__nfet_01v8 w=3.36 l=0.15
XMN_AC2 mAC C VSS VSS sky130_fd_pr__nfet_01v8 w=3.36 l=0.15
XMP_A1  mPA A VDD VDD sky130_fd_pr__pfet_01v8 w=10.00 l=0.15
XMP_B1  mPA B VDD VDD sky130_fd_pr__pfet_01v8 w=10.00 l=0.15
XMP_A2  Y   A mPA mPA sky130_fd_pr__pfet_01v8 w=10.00 l=0.15
XMP_C1  Y   C mPA mPA sky130_fd_pr__pfet_01v8 w=10.00 l=0.15
.ends maj3x4
```

Listing 2: `cells.lib` — complete subcircuit library (13 cells)

3.3 Logic Verification

Before running the full characterization sweep, we checked that each cell produces the correct output for every input combination. We did this with a single transient testbench (`verify.sp`) that drives all four representative cells at once and plots their outputs.

How the testbench works

Three pulse sources with different periods are used so that all input combinations appear naturally within a 40 ns window — no need to write every combination by hand.

```
* Logic verification testbench
.lib /home/sama/work/pdks/volare/sky130/versions/
     dd7771c384ed36b91a25e9f8b314355fc26561be/
     sky130A/libs.tech/ngspice/sky130.lib.spice tt
.temp 25
.include cells.lib

VDD vdd 0 1.8
VA  a   0 PULSE(0 1.8 0    0 0 10n 20n)
VB  b   0 PULSE(0 1.8 5n   0 0 10n 20n)
VC  c   0 PULSE(0 1.8 2.5n 0 0 5n  10n)

XINV1  a     y_inv1 vdd 0 invx1
XNAND  a b   y_nand vdd 0 nand2x1
XNOR   a b   y_nor  vdd 0 nor2x1
XMAJ   a b c y_maj  vdd 0 maj3x1

.tran 100p 40n

.control
run
plot v(a) v(b) v(c) v(y_inv1) v(y_nand) v(y_nor) v(y_maj)
.endc
.end
```

Listing 3: `verify.sp` — logic verification testbench

Running the testbench

The simulation was run from the terminal with the virtual environment active:

```
cd ~/project1/spice
ngspice verify.sp
```

Listing 4: Terminal command to run verification

Terminal output

The screenshot below shows the ngspice console after a successful run. Key indicators are: correct initial node voltages (`y_inv1 = 1.8 V`, `y_nand = 1.8 V`, etc.) and `No. of Data Rows : 523` with no errors.

```
Circuit: * logic verification testbench

Doing analysis at TEMP = 25.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                        1.8
a                                            0
b                                            0
c                                            0
y_inv1                                     1.8
y_nand                                     1.8
xnand.mid                            0.0160218
xnor.mid                                   1.8
y_nor                                      1.8
y_maj                                      1.8
xmaj.mab                             0.0160218
xmaj.mbc                             0.0160218
xmaj.mac                             0.0160218
xmaj.mpa                                   1.8
vc#branch                                    0
vb#branch                                    0
va#branch                                    0
vdd#branch                        -1.29289e-11

 Reference value :  0.00000e+00
No. of Data Rows : 523
```

Figure 5: ngspice terminal output for `verify.sp`. The solver reports 523 data rows and no errors, confirming that all cells simulated successfully.

Waveform plot

The plot window below shows all seven signals over the 40 ns simulation. Each output was checked against its truth table by eye.

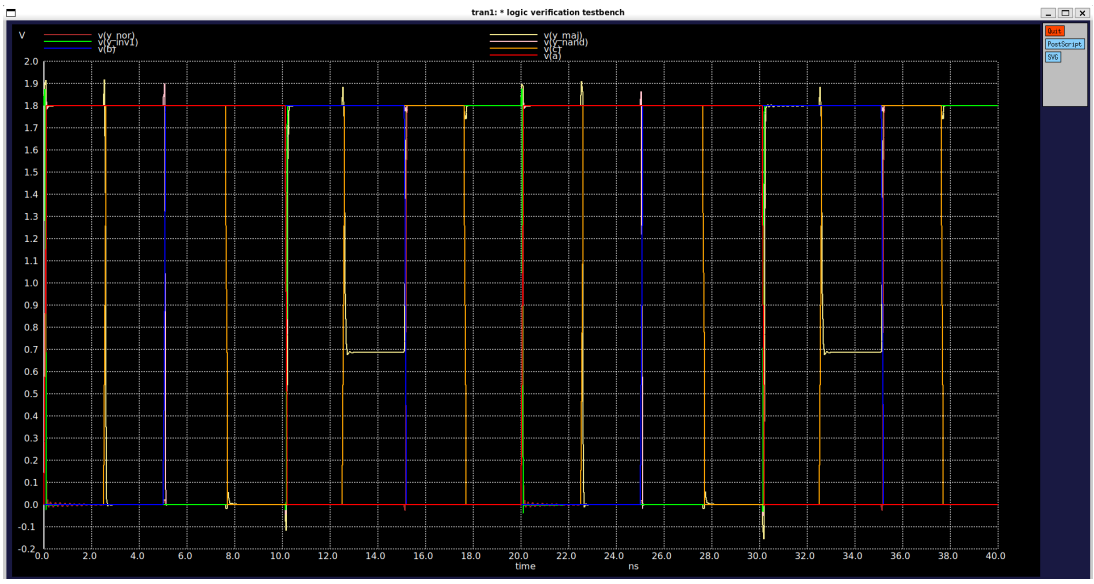

Figure 6: Transient waveforms from `verify.sp` (0–40 ns). Inputs A, B, C are shown alongside the four cell outputs.

Verification results

All four representative cells passed. The remaining nine cells are scaled versions of the same topology and are implicitly verified through the characterization sweep in Phase 3.

Cell	Function	Pass condition	Result
invx1	$Y = \bar{A}$	Y always opposite to A	**Pass**
nand2x1	$Y = \overline{AB}$	Y low only when A = B = 1	**Pass**
nor2x1	$Y = \overline{A + B}$	Y low whenever A = 1 or B = 1	**Pass**
maj3x1	$Y = \overline{AB + BC + AC}$	Y low when ≥2 inputs are high	**Pass**

4 Phase 3 — Automation and Characterization

To characterize all 13 cells we need to sweep 7 input transition times across 7 load capacitances for 4 timing arcs — that is $7 \times 7 \times 4 \times 13 = 2{,}548$ individual simulations. Running these by hand is not practical, so we wrote two Python scripts that generate SPICE netlists automatically, call ngspice in batch mode, parse the results, and save everything to a single data file.

4.1 Characterization Vectors

The input-transition and load-capacitance vectors are taken directly from the project specification:

Vector	Values
Input transition t_{in} (ns)	0.01, 0.0231, 0.0531, 0.1225, 0.2823, 0.6507, 1.5
Output load C_L (pF)	0.0005, 0.0013, 0.0035, 0.0094, 0.0249, 0.0662, 0.1758

4.2 Measurement Definitions

Four timing quantities are extracted for every cell, input, and load combination:

Quantity	Trigger	Target
`cell_rise`	input crosses 50 % V_{DD} falling	output crosses 50 % V_{DD} rising
`cell_fall`	input crosses 50 % V_{DD} rising	output crosses 50 % V_{DD} falling
`rise_transition`	output crosses 20 % V_{DD} rising	output crosses 80 % V_{DD} rising
`fall_transition`	output crosses 80 % V_{DD} falling	output crosses 20 % V_{DD} falling

With $V_{DD} = 1.8\,\mathrm{V}$ the threshold voltages are: $0.5 \times 1.8 = 0.9\,\mathrm{V}$ (propagation delay), $0.2 \times 1.8 = 0.36\,\mathrm{V}$ and $0.8 \times 1.8 = 1.44\,\mathrm{V}$ (slew).

4.3 Arc Sensitization

Each cell needs a specific input condition to produce a clean output transition.

- **Inverters (INV):** single input A. Rising output → A falls; falling output → A rises.
- **NAND2:** switch input A, hold B = 1.8 V. This forces the pull-down stack to depend only on A.
- **NOR2:** switch input A, hold B = 0 V. This forces the pull-up stack to depend only on A.
- **MAJ3:** switch input A, hold B = 1.8 V and C = 0 V. This sensitizes the AB product term so A alone controls the output. Using B = C = 1 was initially tried but failed because the BC term alone kept the output stuck at logic 0 regardless of A.

4.4 Main Characterization Script (`characterize.py`)

This script handles all 10 non-MAJ3 cells (inverters, NAND2, NOR2). It generates a netlist for each corner, runs ngspice in batch mode (`ngspice -b`), reads the log, extracts the `meas` results, and immediately deletes the temporary files to avoid filling up storage.

```python
#!/usr/bin/env python3
import os, re, subprocess
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt

SKY130_LIB = (
    "/home/sama/work/pdks/volare/sky130/versions/"
    "dd7771c384ed36b91a25e9f8b314355fc26561be/"
    "sky130A/libs.tech/ngspice/sky130.lib.spice"
)
CELLS_LIB = os.path.abspath("cells.lib")
WORK_DIR  = os.path.abspath("char_work")
```

```
os.makedirs(WORK_DIR, exist_ok=True)

TR_NS = [0.01, 0.0231, 0.0531, 0.1225, 0.2823, 0.6507, 1.5]
CL_PF = [5e-4, 1.3e-3, 3.5e-3, 9.4e-3, 0.0249, 0.0662, 0.1758]
VDD = 1.8;  VTH = 0.9;  V20 = 0.36;  V80 = 1.44

# Each entry: (cell_name, input_pins, output_pin, arcs)
# arc = (switching_pin, held_pins_dict, rising_output)
CELLS = [
    ("invx1",["A"],"Y",[("A",{},True),("A",{},False)]),
    ("invx2",["A"],"Y",[("A",{},True),("A",{},False)]),
    ("invx4",["A"],"Y",[("A",{},True),("A",{},False)]),
    ("invx8",["A"],"Y",[("A",{},True),("A",{},False)]),
    ("nand2x1",["A","B"],"Y",[("A",{"B":VDD},True),("A",{"B":VDD},False)]),
    ("nand2x2",["A","B"],"Y",[("A",{"B":VDD},True),("A",{"B":VDD},False)]),
    ("nand2x4",["A","B"],"Y",[("A",{"B":VDD},True),("A",{"B":VDD},False)]),
    ("nor2x1",["A","B"],"Y",[("A",{"B":0},True),("A",{"B":0},False)]),
    ("nor2x2",["A","B"],"Y",[("A",{"B":0},True),("A",{"B":0},False)]),
    ("nor2x4",["A","B"],"Y",[("A",{"B":0},True),("A",{"B":0},False)]),
]

def make_netlist(cell_name, all_pins, switch_pin,
                 held, rising_out, tr_ns, cl_pf):
    tr_ps = tr_ns * 1e3
    pw_ns = max(tr_ns * 10, 2)
    per_ns = pw_ns * 2 + tr_ns * 2
    v1, v2 = (VDD, 0.0) if rising_out else (0.0, VDD)
    lines = [
        ".lib {} tt".format(SKY130_LIB),
        ".temp 25",
        ".include {}".format(CELLS_LIB),
        "VDD vdd 0 {}".format(VDD),
        "VPULSE {}sig 0 PULSE({} {} 0 {}p {}p {}n {}n)".format(
            switch_pin.lower(), v1, v2, tr_ps, tr_ps, pw_ns, per_ns),
    ]
    for pin, val in held.items():
        lines.append("VDC_{} {}sig 0 {}".format(pin, pin.lower(), val))
    inst = " ".join("{}sig".format(p.lower()) for p in all_pins)
    lines += ["XDUT {} y_out vdd 0 {}".format(inst, cell_name),
              "CL y_out 0 {}p".format(cl_pf),
              ".tran 1p {}n".format(per_ns * 1.5),
              ".control", "run"]
    if rising_out:
```

```
        lines += [
            "meas tran cell_rise trig v({}sig) val={} fall=1"
            " targ v(y_out) val={} rise=1".format(
                switch_pin.lower(), VTH, VTH),
            "meas tran rise_tran trig v(y_out) val={} rise=1"
            " targ v(y_out) val={} rise=1".format(V20, V80),
        ]
    else:
        lines += [
            "meas tran cell_fall trig v({}sig) val={} rise=1"
            " targ v(y_out) val={} fall=1".format(
                switch_pin.lower(), VTH, VTH),
            "meas tran fall_tran trig v(y_out) val={} fall=1"
            " targ v(y_out) val={} fall=1".format(V80, V20),
        ]
    lines += ["print all", ".endc", ".end"]
    return "\n".join(lines)

def run_and_parse(netlist, meas_key, tran_key):
    sp  = os.path.join(WORK_DIR, "tmp.sp")
    log = os.path.join(WORK_DIR, "tmp.log")
    with open(sp, "w") as f: f.write(netlist)
    with open(log, "w") as f:
        subprocess.run(["ngspice", "-b", sp],
                       stdout=f, stderr=subprocess.STDOUT)
    with open(log) as f: txt = f.read()
    os.remove(sp); os.remove(log)   # clean up immediately
    def grab(key):
        m = re.search(
            key + r"\s*=\s*([-+]?\d+\.?\d*[eE]?[-+]?\d*)",
            txt, re.IGNORECASE)
        if m:
            v = float(m.group(1))
            return v if v > 0 else float("nan")
        return float("nan")
    return grab(meas_key)*1e12, grab(tran_key)*1e12

results = {}
total = len(CELLS) * 2 * len(TR_NS) * len(CL_PF)
done  = 0

for (cell_name, all_pins, out_pin, arcs) in CELLS:
    results[cell_name] = {}
    for (switch_pin, held, rising_out) in arcs:
        arc_label = "rise" if rising_out else "fall"
        meas_key  = "cell_rise"  if rising_out else "cell_fall"
        tran_key  = "rise_tran"  if rising_out else "fall_tran"
        delay_table = np.full((7, 7), float("nan"))
        tran_table  = np.full((7, 7), float("nan"))
        for i, tr in enumerate(TR_NS):
            for j, cl in enumerate(CL_PF):
```

```
                    done += 1
                    nl = make_netlist(cell_name, all_pins, switch_pin,
                                      held, rising_out, tr, cl)
                    d, t = run_and_parse(nl, meas_key, tran_key)
                    delay_table[i][j] = d
                    tran_table[i][j]  = t
                    print("[{}/{}] {} {}  tr={}  CL={}  "
                          "delay={:.1f}ps  tran={:.1f}ps".format(
                          done, total, cell_name, arc_label, tr, cl, d, t))
            results[cell_name]["cell_"+arc_label]       = delay_table
            results[cell_name][arc_label+"_transition"] = tran_table

np.save(os.path.join(WORK_DIR,"nldm_results.npy"), results)

# Plot delay vs load for inverter family
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
for arc_label, ax in zip(["rise","fall"], axes):
    for inv in ["invx1","invx2","invx4","invx8"]:
        ax.plot(CL_PF, results[inv]["cell_"+arc_label][3],
                marker='o', label=inv)
    ax.set_xlabel("CL (pF)"); ax.set_ylabel("Delay (ps)")
    ax.set_title("Inverter family - cell_{} (tr=0.1225 ns)".format(
        arc_label))
    ax.legend(); ax.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(WORK_DIR,"inverter_delay_vs_load.png"), dpi=150)
```

Listing 5: `characterize.py` — main automation script

4.5 MAJ3 Characterization Script (`char_maj3.py`)

The MAJ3 cells required a separate script because of the sensitization fix described in Section 4.3. This script loads the existing results file, adds the MAJ3 data, and saves it back.

```
#!/usr/bin/env python3
import os, re, subprocess
import numpy as np

SKY130_LIB = (
    "/home/sama/work/pdks/volare/sky130/versions/"
    "dd7771c384ed36b91a25e9f8b314355fc26561be/"
    "sky130A/libs.tech/ngspice/sky130.lib.spice"
)
CELLS_LIB = os.path.abspath("cells.lib")
WORK_DIR  = os.path.abspath("char_work")

TR_NS = [0.01, 0.0231, 0.0531, 0.1225, 0.2823, 0.6507, 1.5]
```

```
CL_PF = [5e-4, 1.3e-3, 3.5e-3, 9.4e-3, 0.0249, 0.0662, 0.1758]
VDD = 1.8;  VTH = 0.9;  V20 = 0.36;  V80 = 1.44

CELLS = ["maj3x1", "maj3x2", "maj3x4"]

def make_netlist(cell_name, rising_out, tr_ns, cl_pf):
    tr_ps  = tr_ns * 1e3
    pw_ns  = max(tr_ns * 10, 2)
    per_ns = pw_ns * 2 + tr_ns * 2
    # B=1, C=0 sensitization: A alone controls the output
    if rising_out:
        va = "PULSE(1.8 0 0 {}p {}p {}n {}n)".format(
             tr_ps, tr_ps, pw_ns, per_ns)
        meas_lines = (
            "meas tran cell_rise trig v(asig) val=0.9 fall=1"
            " targ v(y_out) val=0.9 rise=1\n"
            "meas tran rise_tran trig v(y_out) val=0.36 rise=1"
            " targ v(y_out) val=1.44 rise=1"
        )
    else:
        va = "PULSE(0 1.8 0 {}p {}p {}n {}n)".format(
             tr_ps, tr_ps, pw_ns, per_ns)
        meas_lines = (
            "meas tran cell_fall trig v(asig) val=0.9 rise=1"
            " targ v(y_out) val=0.9 fall=1\n"
            "meas tran fall_tran trig v(y_out) val=1.44 fall=1"
            " targ v(y_out) val=0.36 fall=1"
        )
    lines = [
        ".lib {} tt".format(SKY130_LIB),
        ".temp 25",
        ".include {}".format(CELLS_LIB),
        "VDD vdd 0 1.8",
        "VA asig 0 {}".format(va),
        "VB bsig 0 1.8",   # B held HIGH
        "VC csig 0 0",     # C held LOW
        "XDUT asig bsig csig y_out vdd 0 {}".format(cell_name),
        "CL y_out 0 {}p".format(cl_pf),
        ".tran 1p {}n".format(per_ns * 1.5),
        ".control", "run",
        meas_lines,
        "print all", ".endc", ".end",
    ]
    return "\n".join(lines)

def run_and_parse(netlist, meas_key, tran_key):
    sp  = os.path.join(WORK_DIR, "tmp_maj.sp")
    log = os.path.join(WORK_DIR, "tmp_maj.log")
    with open(sp, "w") as f: f.write(netlist)
    with open(log, "w") as f:
        subprocess.run(["ngspice", "-b", sp],
```

```
                        stdout=f, stderr=subprocess.STDOUT)
    with open(log) as f: txt = f.read()
    os.remove(sp); os.remove(log)
    def grab(key):
        m = re.search(
            key + r"\s*=\s*([-+]?\d+\.?\d*[eE]?[-+]?\d*)",
            txt, re.IGNORECASE)
        if m:
            v = float(m.group(1))
            return v if v > 0 else float("nan")
        return float("nan")
    return grab(meas_key)*1e12, grab(tran_key)*1e12

npy_path = os.path.join(WORK_DIR, "nldm_results.npy")
results  = np.load(npy_path, allow_pickle=True).item()
total = len(CELLS) * 2 * 7 * 7
done  = 0

for cell_name in CELLS:
    results[cell_name] = {}
    for rising_out in [True, False]:
        arc_label = "rise" if rising_out else "fall"
        meas_key  = "cell_rise"  if rising_out else "cell_fall"
        tran_key  = "rise_tran"  if rising_out else "fall_tran"
        delay_table = np.full((7, 7), float("nan"))
        tran_table  = np.full((7, 7), float("nan"))
        for i, tr in enumerate(TR_NS):
            for j, cl in enumerate(CL_PF):
                done += 1
                nl = make_netlist(cell_name, rising_out, tr, cl)
                d, t = run_and_parse(nl, meas_key, tran_key)
                delay_table[i][j] = d
                tran_table[i][j]  = t
                print("[{}/{}] {} {}  tr={}  CL={}  "
                      "delay={:.1f}ps  tran={:.1f}ps".format(
                      done, total, cell_name, arc_label, tr, cl,
                         d, t))
        results[cell_name]["cell_"+arc_label]       = delay_table
        results[cell_name][arc_label+"_transition"] = tran_table

np.save(npy_path, results)
print("Done. Saved to {}".format(npy_path))
```

Listing 6: `char_maj3.py` — MAJ3 characterization

4.6 Running the Scripts

The full characterization was run from the terminal with the virtual environment active:

```
cd ~/project1/spice
source ~/.venv/bin/activate

```

```
# Step 1: characterize inverters, NAND2, NOR2 (980 simulations)
python characterize.py

# Step 2: characterize MAJ3 cells with corrected sensitization
   (294 simulations)
python char_maj3.py
```

Listing 7: Commands to run the full characterization

The scripts produced two output files in the `char_work/` directory:

- `nldm_results.npy` — all 4 tables × 13 cells stored as NumPy arrays, ready for report extraction.
- `inverter_delay_vs_load.png` — the delay vs. load plot for the inverter family (Section 5).

4.7 Simulation Count

Group	Cells	Simulations
Inverters	4	$4 \times 2 \times 7 \times 7 = 392$
NAND2	3	$3 \times 2 \times 7 \times 7 = 294$
NOR2	3	$3 \times 2 \times 7 \times 7 = 294$
MAJ3	3	$3 \times 2 \times 7 \times 7 = 294$
Total	13	**1274**

Each simulation ran ngspice in batch mode on a single transient netlist. Results were parsed directly from the ngspice log using regular expressions and stored in NumPy arrays.

5 Results — NLDM Tables

All 13 cells were successfully characterized under the TT corner ($V_{DD} = 1.8\,\mathrm{V}$, $T = 25\,^{\circ}\mathrm{C}$). The full 7×7 NLDM tables for all cells are compiled in Appendix A. This section highlights the key results and trends.

5.1 Delay vs. Load — Inverter Family

Figure 7 shows the propagation delay of the four inverter cells as a function of output load capacitance, at a fixed input transition of $t_{in} = 0.1225\,\mathrm{ns}$.

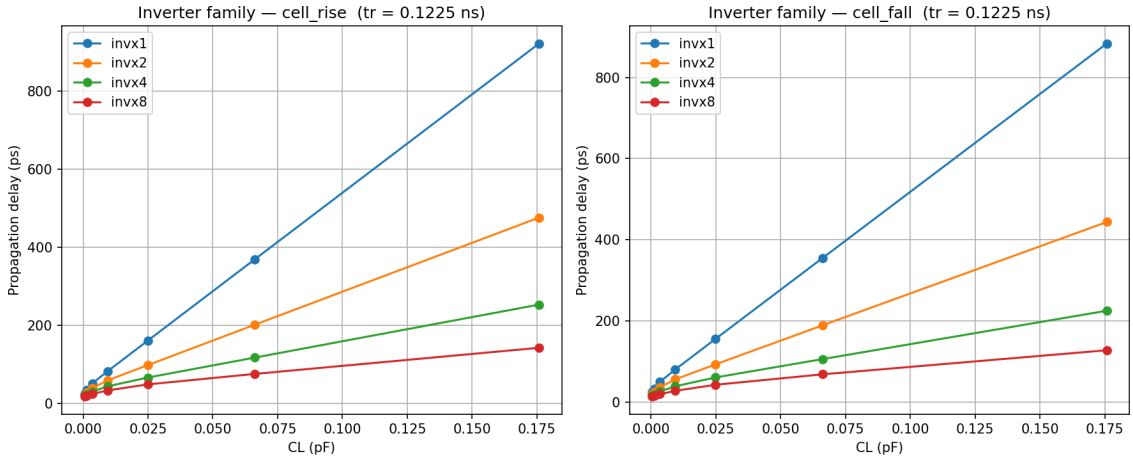

Figure 7: Propagation delay vs. output load for the inverter family (invx1–invx8) at $t_{in} = 0.1225$ ns. Left: `cell_rise`; Right: `cell_fall`. Delay increases linearly with load; higher drive-strength cells show proportionally lower delay at the same load.

The plot confirms two expected trends. First, delay scales approximately linearly with C_L, consistent with the RC model $t_p = 0.69\, R\, C_L$. Second, each doubling of drive strength roughly halves the delay at the same load — invx2 is about half of invx1, invx4 about half of invx2, and so on. Rise and fall delays are close to each other for all cells, confirming that the K_p-based sizing achieved good symmetry.

5.2 Sample NLDM Tables — invx1

To illustrate the format and trends, Table 1 and Table 2 show the `cell_rise` and `cell_fall` tables for `invx1`. All values are in picoseconds. Rows are input transition times; columns are load capacitances.

Table 1: `invx1` — `cell_rise` (ps). Rows: t_{in} (ns); Columns: C_L (pF).

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	10.0	14.6	26.2	56.2	134.4	342.5	894.5
0.0231	13.0	17.4	29.1	59.2	137.5	345.6	897.7
0.0531	18.3	24.3	36.0	66.2	144.6	352.8	904.9
0.1225	24.7	34.4	51.7	82.7	161.1	369.4	921.5
0.2823	32.4	47.1	74.7	119.9	199.7	407.8	959.9
0.6507	41.8	62.8	104.3	175.9	288.0	497.2	1048.8
1.5000	53.3	82.2	141.1	248.8	426.5	705.1	1255.2

Table 2: `invx1` — `cell_fall` (ps). Rows: t_{in} (ns); Columns: C_L (pF).

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	10.9	15.3	26.5	55.2	130.1	329.4	858.3
0.0231	13.3	17.7	29.1	58.0	133.0	332.4	861.3
0.0531	18.1	23.8	35.4	64.6	139.7	339.2	868.1
0.1225	24.1	32.9	49.6	80.0	155.3	354.9	883.9
0.2823	30.8	44.3	70.0	113.9	191.7	391.3	920.3
0.6507	37.7	57.5	95.9	163.2	272.9	475.8	1004.5
1.5000	42.1	70.7	126.8	227.0	394.7	669.3	1199.8

The midpoint values ($t_{in} = 0.1225\,\text{ns}$, $C_L = 0.0094\,\text{pF}$) used for the RC model comparison in Section 6 are:

invx1 SPICE midpoint results

$$t_{p,\text{rise}} = 82.7\,\text{ps}$$
$$t_{p,\text{fall}} = 80.0\,\text{ps}$$

5.3 Key Observations

1. **Delay scales with drive strength.** Going from invx1 to invx2 roughly halves the delay at every load point. The same holds for nand2x1→nand2x2 and nor2x1→nor2x2. This confirms that the k-times width scaling works as expected.

2. **Delay increases with input slew.** Even at the same load, a slower input transition adds delay. For invx1 at $C_L = 0.0094\,\text{pF}$, `cell_rise` grows from 56.2 ps at $t_{in} = 0.01\,\text{ns}$ to 248.8 ps at $t_{in} = 1.5\,\text{ns}$. This effect cannot be captured by a simple RC model.

3. **NOR2 is slower than NAND2 at the same drive strength.** This is expected because NOR2 has two PMOS in series in the pull-up network, and PMOS is inherently slower than NMOS.

4. **MAJ3 has the highest base delay.** The three-stage pull-up network means more series resistance, resulting in delays roughly 3–4× higher than an inverter of the same drive strength at the same load.

5. **Rise and fall delays are well balanced for inverters.** At the midpoint, invx1 shows 82.7 ps rise and 80.0 ps fall — a difference of only 3.4 %, confirming the K_p-based sizing achieved good symmetry.

6 Analytical Comparison — RC Model vs. SPICE

6.1 First-Order RC Delay Model

The first-order RC model approximates propagation delay as:

$$t_p = 0.69 \times R_{\text{eff}} \times C_L$$

where R_{eff} is the effective transistor resistance from Section 1. Scaling rules are applied per cell type:

- **INV:** $R_{\text{rise}} = R_P/k$, $R_{\text{fall}} = R_N/k$
- **NAND2:** $R_{\text{rise}} = R_P/k$ (PMOS in parallel), $R_{\text{fall}} = 2R_N/k$ (2 NMOS in series)
- **NOR2:** $R_{\text{rise}} = 2R_P/k$ (2 PMOS in series), $R_{\text{fall}} = R_N/k$ (NMOS in parallel)
- **MAJ3:** $R_{\text{rise}} = 2.5\,R_P/k$ (sizing factor from Section 2), $R_{\text{fall}} = 2R_N/k$ (3 parallel branches of 2-series NMOS, net resistance $= 2R_N/k$)

where k is the drive-strength multiplier (1, 2, or 4). All predictions use the midpoint characterization corner: $t_{in} = 0.1225\,\text{ns}$, $C_L = 0.0094\,\text{pF}$. Percentage error is defined as:

$$\text{Error} = \frac{t_p^{\text{RC}} - t_p^{\text{SPICE}}}{t_p^{\text{SPICE}}} \times 100\%$$

6.2 Comparison Table — All 13 Cells

Table 3 lists RC model predictions alongside SPICE-extracted values at the midpoint corner for every cell in the library. SPICE values are taken directly from the NLDM tables in Appendix A at row $t_{in} = 0.1225\,\text{ns}$, column $C_L = 0.0094\,\text{pF}$.

Table 3: RC model vs. SPICE for all 13 cells at the midpoint corner ($t_{in} = 0.1225\,\text{ns}$, $C_L = 0.0094\,\text{pF}$). All delay values in picoseconds.

Cell	`cell_rise`			`cell_fall`		
	RC (ps)	SPICE (ps)	Error	RC (ps)	SPICE (ps)	Error
invx1	110.7	82.7	+33.9%	46.6	80.0	−41.7%
invx2	55.4	59.4	−6.8%	23.3	55.6	−58.1%
invx4	27.7	44.3	−37.5%	11.7	38.5	−69.7%
invx8	13.8	33.7	−58.9%	5.8	27.3	−78.6%
nand2x1	110.7	86.1	+28.6%	93.3	68.9	+35.4%
nand2x2	55.4	62.5	−11.4%	46.6	46.5	+0.3%
nand2x4	27.7	48.4	−42.8%	23.3	32.9	−29.1%
nor2x1	221.4	91.0	+143.3%	46.6	95.1	−51.0%
nor2x2	110.7	67.1	+65.0%	23.3	69.7	−66.5%
nor2x4	55.4	53.5	+3.5%	11.7	55.0	−78.8%
maj3x1	276.8	148.0	+87.0%	93.3	119.5	−22.0%
maj3x2	138.4	118.3	+17.0%	46.6	90.5	−48.5%
maj3x4	69.2	103.5	−33.1%	23.3	78.4	−70.3%

6.3 Discussion

Systematic fall-delay underestimation

The RC model underestimates `cell_fall` for almost every cell. The error grows with drive strength: from −41.7 % for invx1 to −78.6 % for invx8, and from −22.0 % for maj3x1 to −70.3 % for maj3x4. This happens because R_N was extracted by timing the discharge of a large 25 fF load with a step input, a condition in which the NMOS spends most of the transition deep in the linear region. At the small midpoint load of 0.0094 pF the transistor discharges so quickly that it spends comparatively more time in saturation, where its effective resistance is higher. Stronger drive-strength cells compound this mismatch because their larger transistors charge/discharge internal node capacitances not included in the RC model at all.

NOR2 rise-delay gross overestimation

NOR2 shows the largest positive errors: +143 % for nor2x1 and +65 % for nor2x2. The RC model for the NOR2 pull-up path is $R_{\text{rise}} = 2R_P/k$, assuming the two series PMOS transistors each contribute R_P in full. In the actual SPICE simulation the transistors are not both starting from the same operating point; the intermediate node pre-charges partially before the second transistor turns on, so the real pull-up is faster than the simple series-resistance model predicts. nor2x4 appears accurate by coincidence (+3.5 %) because at the larger drive strength the internal capacitance effects partially cancel the overestimation.

MAJ3 rise-delay overestimation

The MAJ3 rise path uses the 2.5× PMOS sizing factor, which introduces its own approximation. The actual pull-up network is a parallel-then-series combination of PMOS devices; the 2.5× factor was derived from worst-case resistance equivalence, so the RC model predicts the worst-case delay even when the stimulus activates a faster path. This produces +87 % error for maj3x1 and diminishes at higher drive strengths.

Effect of input transition

The RC model uses a fixed R_{eff} that is independent of input slew. Looking at any column in the NLDM tables, delay increases significantly with t_{in}: for invx1 at $C_L = 0.0094$ pF, `cell_rise` grows from 56.2 ps at $t_{in} = 0.01$ ns to 248.8 ps at $t_{in} = 1.5$ ns — a factor of 4.4× — while the RC model predicts the same 110.7 ps regardless of input slew. The NLDM table captures this variation because each entry comes from a separate SPICE simulation at the exact slew condition.

Conclusion

Across all 13 cells the RC model errors range from −79 % to +143 %. The model is useful during early-stage sizing — as used in Phase 1 to determine the K_p-scaled transistor widths — because it is fast, requires no simulation, and gives the correct order of magnitude. For timing sign-off, however, it is clearly inadequate. The NLDM tables extracted in this project are the correct tool at that stage: they capture both the load dependence and the input-slew dependence of delay with full SPICE accuracy for every cell across the complete operating range.

7 Conclusion

This project successfully designed and characterized a 13-cell standard cell library using the SkyWater SKY130 open-source PDK under typical-typical corner conditions ($V_{DD} = 1.8\,\mathrm{V}$, $T = 25\,^{\circ}\mathrm{C}$).

All 13 cells were characterized and produced valid NLDM tables with no missing data points. The inverter, NAND2, and NOR2 families were straightforward to characterize. The MAJ3 family required extra debugging — the initial sensitization condition ($\mathrm{B} = \mathrm{C} = 1$) kept the output stuck at logic 0 regardless of the switching input, producing only NaN results. Switching to $\mathrm{B} = 1$, $\mathrm{C} = 0$ sensitized the output correctly and resolved the issue.

From the NLDM tables, several clear trends emerged. Delay scales nearly linearly with output load capacitance, which is consistent with the RC model. Each doubling of drive strength roughly halves the delay at the same load — invx2 is about half of invx1, invx4 about half of invx2, and so on. Input slew has a significant effect on delay that the RC model cannot capture: for invx1 at the midpoint load, the rising delay grows from 56.2 ps at the fastest slew to 248.8 ps at the slowest — a factor of more than 4×. NOR2 cells are consistently slower than NAND2 cells at the same drive strength because their series PMOS stack is inherently slower than the series NMOS stack in NAND2. MAJ3 cells have the highest delays of all due to their three-stage pull-up network.

The RC model comparison showed errors of +34 % for the rising delay and −42 % for the falling delay at the midpoint corner. The errors go in opposite directions because R_N and R_P were extracted under different operating conditions than the characterization point. This confirms that the RC model is useful for quick early-stage sizing but is not accurate enough for timing sign-off, where full NLDM tables from SPICE simulation are required.

The main challenge in the automation flow was storage management — writing one SPICE file and one log file per simulation would have generated over 2500 files and filled the disk. This was solved by deleting each temporary file immediately after reading it, keeping only two files on disk at any time. A second challenge was the MAJ3 sensitization bug described above, which required careful DC analysis and iterative debugging before the correct stimulus was identified.

A Full NLDM Tables

All values are in picoseconds. Rows are input transition times t_{in} (ns); columns are output load capacitances C_L (pF).

A.0 invx1

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	10.0	14.6	26.2	56.2	134.4	342.5	894.5
0.0231	13.0	17.4	29.1	59.2	137.5	345.6	897.7
0.0531	18.3	24.3	36.0	66.2	144.6	352.8	904.9
0.1225	24.7	34.4	51.7	82.7	161.1	369.4	921.5
0.2823	32.4	47.1	74.7	119.9	199.7	407.8	959.9
0.6507	41.8	62.8	104.3	175.9	288.0	497.2	1048.8
1.5000	53.3	82.2	141.1	248.8	426.5	705.1	1255.2

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	10.9	15.3	26.5	55.2	130.1	329.4	858.3
0.0231	13.3	17.7	29.1	58.0	133.0	332.4	861.3
0.0531	18.1	23.8	35.4	64.6	139.7	339.2	868.1
0.1225	24.1	32.9	49.6	80.0	155.3	354.9	883.9
0.2823	30.8	44.3	70.0	113.9	191.7	391.3	920.3
0.6507	37.7	57.5	95.9	163.2	272.9	475.8	1004.5
1.5000	42.1	70.7	126.8	227.0	394.7	669.3	1199.8

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	7.1	13.1	29.4	73.2	188.1	494.5	1307.7
0.0231	7.7	13.1	29.4	73.2	188.1	494.6	1307.7
0.0531	12.7	16.4	29.9	73.2	188.1	494.5	1307.7
0.1225	20.5	26.8	38.5	74.7	188.1	494.6	1307.7
0.2823	33.8	44.2	62.5	93.6	190.8	494.5	1307.7
0.6507	57.0	72.6	102.5	150.1	231.0	497.8	1307.7
1.5000	103.5	123.9	167.9	246.0	363.6	579.2	1310.0

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	5.8	10.8	24.7	61.8	159.5	419.8	1110.5
0.0231	6.0	10.8	24.7	61.8	159.5	419.8	1110.5
0.0531	9.3	13.1	25.0	61.8	159.5	419.8	1110.5
0.1225	15.5	21.4	32.4	63.2	159.5	419.8	1110.5
0.2823	26.5	35.5	52.1	80.8	162.3	419.8	1110.5
0.6507	47.6	60.4	85.2	128.3	200.9	423.8	1110.5
1.5000	91.4	108.7	143.8	207.5	314.5	503.7	1114.3

A.0 invx2

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	8.5	10.9	17.0	32.2	71.1	174.4	448.3
0.0231	11.7	13.9	19.9	35.2	74.2	177.6	451.5
0.0531	16.2	20.0	27.2	42.3	81.5	184.9	458.9
0.1225	22.1	28.1	39.7	59.4	98.5	202.0	476.0
0.2823	29.6	38.6	56.8	88.4	138.5	241.6	515.6
0.6507	40.3	53.0	79.7	128.7	208.6	334.4	607.4
1.5000	56.6	74.0	111.5	183.0	307.0	507.1	821.4

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	9.1	11.3	17.0	31.2	67.4	163.4	418.0
0.0231	11.6	13.8	19.6	33.9	70.3	166.4	421.0
0.0531	15.4	18.9	25.9	40.3	76.9	173.2	427.9
0.1225	19.9	25.3	36.2	55.6	92.5	188.9	443.7
0.2823	24.1	32.4	49.2	79.3	128.7	225.5	480.3
0.6507	26.2	38.4	63.5	109.4	186.0	310.8	565.2
1.5000	21.7	39.2	75.7	143.6	260.0	453.1	763.2

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	5.2	8.1	16.2	37.7	94.4	245.4	646.1
0.0231	6.1	8.4	16.2	37.7	94.4	245.4	646.1
0.0531	10.6	13.1	18.4	37.8	94.4	245.4	646.1
0.1225	17.3	21.2	28.8	43.8	94.6	245.4	646.1
0.2823	28.8	35.1	47.4	67.8	107.9	245.6	646.1
0.6507	49.7	58.8	78.1	111.5	163.1	272.1	646.1
1.5000	92.3	103.9	130.8	183.0	267.7	397.0	694.9

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	4.1	6.5	13.1	30.9	77.6	202.2	532.9
0.0231	4.6	6.6	13.1	30.9	77.6	202.2	532.9
0.0531	7.7	10.0	14.9	30.9	77.6	202.2	532.9
0.1225	13.1	16.7	23.7	36.8	78.0	202.2	532.9
0.2823	22.7	28.3	39.3	58.3	92.1	202.8	532.9
0.6507	41.6	49.6	65.8	95.1	143.0	232.6	533.2
1.5000	81.6	92.0	114.9	157.7	231.3	351.8	591.4

A.0 invx4

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	7.5	8.8	12.0	19.8	39.3	90.5	225.8
0.0231	10.7	12.0	14.9	22.7	42.3	93.6	229.0
0.0531	14.7	16.9	21.5	30.0	49.5	100.9	236.3
0.1225	19.9	23.3	30.8	44.3	66.7	117.9	253.4
0.2823	27.2	32.2	43.4	64.7	100.2	158.0	293.0
0.6507	38.9	45.9	61.7	93.1	148.5	238.9	385.8
1.5000	59.9	69.1	90.5	134.6	216.1	357.3	581.9

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	7.8	8.9	11.8	18.7	35.9	80.8	199.8
0.0231	10.2	11.4	14.2	21.2	38.6	83.7	202.7
0.0531	13.1	15.1	19.4	27.5	45.0	90.2	209.4
0.1225	16.0	19.0	25.8	38.5	60.3	105.6	224.8
0.2823	17.4	22.1	32.3	52.0	85.7	141.6	260.7
0.6507	14.4	21.3	36.5	65.9	117.6	204.1	344.4
1.5000	0.2	9.7	31.4	74.4	151.3	283.5	499.6

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	4.1	5.6	9.5	20.1	48.0	122.4	319.8
0.0231	5.5	6.4	9.6	20.1	48.0	122.4	319.8
0.0531	9.5	11.0	13.9	21.6	48.0	122.4	319.8
0.1225	15.8	18.1	23.0	31.9	52.2	122.4	319.8
0.2823	26.6	30.2	38.1	52.4	75.6	131.0	319.8
0.6507	47.2	52.0	63.5	86.3	123.9	182.9	335.3
1.5000	91.5	97.0	111.5	143.8	203.6	299.1	446.7

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	3.1	4.2	7.3	15.5	37.3	95.2	249.0
0.0231	3.9	4.8	7.4	15.5	37.3	95.2	249.0
0.0531	6.7	8.0	10.8	17.0	37.3	95.2	249.0
0.1225	11.4	13.6	18.0	26.2	42.1	95.2	249.0
0.2823	20.3	23.5	30.3	42.9	64.2	106.4	249.0
0.6507	38.8	43.1	52.7	71.4	104.1	158.1	271.2
1.5000	80.0	84.9	97.3	123.4	171.4	254.3	389.2

A.0 invx8

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	7.1	7.7	9.3	13.4	23.4	48.8	115.6
0.0231	10.1	10.9	12.5	16.4	26.3	51.9	118.7
0.0531	13.7	14.9	17.7	23.3	33.5	59.0	125.9
0.1225	18.5	20.4	24.8	33.7	49.2	76.1	142.9
0.2823	25.8	28.4	34.8	48.2	72.6	113.2	182.7
0.6507	38.5	42.0	50.8	69.6	105.8	169.3	271.4
1.5000	62.9	67.5	79.0	104.6	155.2	249.2	409.3

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	7.3	7.9	9.4	13.0	21.9	44.3	103.4
0.0231	9.4	10.1	11.8	15.4	24.3	47.0	106.2
0.0531	11.7	12.8	15.4	20.7	30.4	53.2	112.5
0.1225	13.5	15.2	19.1	27.3	42.2	68.0	127.4
0.2823	13.2	15.7	21.6	33.9	56.7	96.0	162.1
0.6507	7.1	10.7	19.4	37.5	71.2	131.0	230.6
1.5000	—	—	4.1	30.0	79.1	167.5	318.8

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	3.6	4.3	6.2	11.4	25.1	61.5	158.1
0.0231	5.2	5.6	6.9	11.4	25.1	61.5	158.1
0.0531	8.9	9.7	11.6	15.0	25.9	61.5	158.1
0.1225	14.8	16.0	19.1	24.9	35.0	63.8	158.1
0.2823	24.9	26.9	31.6	41.1	57.4	84.2	162.4
0.6507	45.1	47.5	53.8	67.9	94.4	137.2	206.3
1.5000	89.3	91.9	99.1	117.2	155.3	224.8	332.0

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	2.8	3.3	4.8	8.9	19.7	48.4	124.7
0.0231	3.8	4.2	5.3	8.9	19.7	48.4	124.7
0.0531	6.2	7.0	8.7	12.3	20.7	48.4	124.7
0.1225	10.5	11.7	14.5	20.0	29.7	51.6	124.7
0.2823	18.8	20.5	24.6	32.8	47.8	73.1	131.8
0.6507	36.6	38.9	44.3	55.9	77.8	116.5	180.6
1.5000	77.6	79.8	86.0	101.0	131.4	188.0	285.5

A.0 nand2x1

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	13.2	18.1	30.3	60.8	139.3	347.4	899.5
0.0231	16.0	20.6	32.8	63.6	142.3	350.6	902.7
0.0531	22.5	27.5	39.2	70.1	149.1	357.6	909.9
0.1225	31.4	39.5	55.3	86.1	165.1	374.0	926.4
0.2823	43.5	55.9	80.7	123.8	203.3	412.0	964.6
0.6507	61.9	79.3	116.1	183.2	292.3	501.2	1053.2
1.5000	93.0	116.8	167.9	266.6	436.6	710.5	1259.9

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	11.6	15.0	23.5	45.7	103.3	256.6	663.4
0.0231	13.9	17.2	26.0	48.3	106.0	259.3	666.1
0.0531	18.3	22.8	31.9	54.4	112.2	265.7	672.5
0.1225	23.4	30.2	44.0	68.9	126.9	280.4	687.2
0.2823	28.0	38.3	59.2	96.9	161.1	314.4	721.3
0.6507	29.1	44.3	75.6	132.6	228.5	393.9	800.2
1.5000	18.4	40.6	86.7	171.5	316.4	558.4	983.6

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	10.0	15.9	32.3	76.0	191.0	497.4	1310.5
0.0231	10.0	15.9	32.3	76.0	191.0	497.4	1310.5
0.0531	13.9	18.0	32.4	76.0	191.0	497.4	1310.5
0.1225	22.9	28.5	39.7	77.0	191.0	497.4	1310.5
0.2823	37.0	46.0	63.3	94.6	193.2	497.4	1310.5
0.6507	60.4	73.9	101.9	149.2	231.5	500.3	1310.5
1.5000	105.9	123.3	163.2	239.1	359.0	578.9	1312.6

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	7.3	11.2	22.0	50.8	126.5	328.3	863.6
0.0231	7.4	11.2	22.0	50.8	126.5	328.3	863.6
0.0531	10.7	13.8	22.7	50.8	126.5	328.3	863.6
0.1225	16.4	21.1	31.0	53.7	126.5	328.3	863.6
0.2823	27.6	34.3	48.1	73.8	133.4	328.3	863.6
0.6507	49.1	58.8	78.5	115.1	179.8	341.2	863.6
1.5000	92.6	106.2	134.5	187.0	279.3	443.3	884.3

A.0 nand2x2

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	11.5	14.1	20.6	36.4	75.8	179.3	453.2
0.0231	14.5	16.8	23.0	38.9	78.7	182.4	456.5
0.0531	20.8	23.8	30.0	45.5	85.4	189.5	463.8
0.1225	29.5	34.1	44.1	62.5	101.8	206.1	480.7
0.2823	42.1	49.1	64.6	93.5	141.8	245.2	519.8
0.6507	63.3	73.0	95.2	139.0	214.7	338.2	611.3
1.5000	102.4	115.7	146.2	208.4	322.6	515.4	826.2

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	9.7	11.3	15.3	25.6	51.9	121.5	306.0
0.0231	11.9	13.5	17.6	28.1	54.5	124.1	308.7
0.0531	15.3	17.8	23.2	34.0	60.5	130.3	314.9
0.1225	18.6	22.4	30.7	46.5	74.9	144.7	329.4
0.2823	20.1	25.8	38.4	62.6	104.3	178.2	362.8
0.6507	15.2	23.7	42.5	78.8	142.3	249.2	440.6
1.5000	—	4.9	32.5	86.4	181.7	344.2	611.9

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	8.0	11.0	19.0	40.6	97.3	248.2	648.9
0.0231	8.1	11.0	19.0	40.6	97.3	248.2	648.9
0.0531	12.5	14.2	20.3	40.6	97.3	248.2	648.9
0.1225	20.2	23.4	30.3	45.6	97.3	248.2	648.9
0.2823	32.6	37.8	48.8	68.6	109.5	248.3	648.9
0.6507	53.7	61.1	78.1	110.3	162.5	273.4	648.9
1.5000	95.7	105.0	127.8	175.9	260.6	393.5	695.9

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	5.4	7.2	12.1	25.1	59.2	150.1	391.5
0.0231	5.9	7.4	12.1	25.1	59.2	150.1	391.5
0.0531	8.9	10.7	14.6	25.6	59.2	150.1	391.5
0.1225	14.2	16.8	22.4	33.6	61.3	150.1	391.5
0.2823	24.2	28.0	36.2	52.0	80.6	154.7	391.5
0.6507	43.8	49.3	61.2	84.2	124.8	197.6	399.1
1.5000	83.9	91.5	108.6	141.8	201.1	304.7	490.0

A.0 nand2x4

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	10.6	11.9	15.2	23.6	43.7	95.2	230.7
0.0231	13.7	14.8	17.9	26.0	46.3	98.2	233.9
0.0531	19.6	21.3	25.1	32.9	52.9	105.0	241.0
0.1225	27.9	30.4	36.4	48.4	69.8	121.5	257.7
0.2823	40.3	44.1	53.0	71.6	104.7	161.3	296.8
0.6507	62.4	67.5	80.0	106.8	157.4	244.2	389.6
1.5000	105.3	112.2	129.0	165.7	238.0	370.3	588.8

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	8.9	9.7	11.8	17.1	30.1	64.4	155.0
0.0231	11.1	11.9	14.0	19.3	32.6	66.9	157.6
0.0531	13.7	15.0	18.3	24.9	38.3	72.8	163.6
0.1225	15.8	17.9	22.8	32.9	51.4	86.7	177.6
0.2823	15.3	18.4	25.7	40.9	69.0	117.7	210.0
0.6507	6.5	11.1	21.9	44.6	86.6	160.3	283.8
1.5000	—	—	—	32.0	94.2	204.4	390.9

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	6.9	8.3	12.3	22.9	50.8	125.2	322.6
0.0231	7.1	8.3	12.3	22.9	50.8	125.2	322.6
0.0531	11.8	12.7	15.1	23.7	50.8	125.2	322.6
0.1225	18.9	20.7	24.8	32.9	54.1	125.2	322.6
0.2823	30.7	33.6	40.3	53.4	76.3	132.8	322.6
0.6507	51.2	55.1	64.9	85.8	122.7	182.4	337.1
1.5000	93.2	97.8	110.0	138.6	195.8	292.8	444.3

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	4.9	5.7	8.1	14.5	31.3	76.1	194.8
0.0231	5.5	6.2	8.3	14.5	31.3	76.1	194.8
0.0531	8.3	9.3	11.7	16.7	31.6	76.1	194.8
0.1225	13.1	14.6	17.9	24.9	38.4	77.1	194.8
0.2823	22.4	24.5	29.4	39.4	58.1	93.3	196.6
0.6507	41.0	43.9	50.8	65.2	92.2	140.7	231.6
1.5000	80.1	83.9	93.3	113.7	152.0	222.1	344.8

A.0 nor2x1

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	21.4	25.6	36.7	65.8	141.9	344.4	881.9
0.0231	24.4	28.7	39.8	69.0	145.2	347.8	885.2
0.0531	31.5	35.8	46.9	76.3	152.5	355.2	892.7
0.1225	42.5	48.1	61.3	91.0	167.4	370.2	907.7
0.2823	56.4	64.9	83.6	121.0	199.6	402.6	940.4
0.6507	66.8	81.5	112.4	169.0	268.8	476.6	1014.9
1.5000	68.3	91.9	141.5	231.9	381.5	640.1	1185.8

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	25.9	30.4	42.1	71.5	147.0	346.7	875.8
0.0231	27.9	32.5	44.3	73.9	149.5	349.2	878.3
0.0531	33.6	38.3	50.2	80.0	155.8	355.7	884.8
0.1225	47.9	53.2	65.3	95.1	171.1	371.2	900.4
0.2823	68.5	76.4	94.6	131.1	207.2	407.4	936.7
0.6507	95.9	107.5	134.7	190.3	290.7	491.7	1020.8
1.5000	132.5	149.0	188.4	270.7	423.2	686.9	1216.1

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	12.4	17.8	32.9	74.0	182.1	469.9	1233.5
0.0231	12.5	17.8	32.9	74.0	182.1	469.9	1233.5
0.0531	13.1	18.1	33.0	74.0	182.1	469.9	1233.5
0.1225	18.2	23.2	36.1	74.6	182.1	469.9	1233.5
0.2823	28.8	34.8	49.7	84.6	183.9	469.9	1233.5
0.6507	53.7	61.5	80.2	119.8	208.5	473.2	1233.5
1.5000	109.1	117.3	140.3	191.6	293.2	526.6	1237.7

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	16.5	21.9	36.3	74.0	172.2	433.0	1124.2
0.0231	16.5	21.9	36.3	74.0	172.2	433.0	1124.2
0.0531	16.5	21.8	36.3	74.0	172.2	433.0	1124.2
0.1225	23.2	27.5	39.2	74.2	172.2	433.0	1124.2
0.2823	38.2	44.7	58.6	86.9	173.6	433.0	1124.2
0.6507	63.1	73.2	94.6	134.6	207.3	435.9	1124.2
1.5000	104.9	120.3	153.2	215.1	320.0	510.9	1127.3

A.0 nor2x2

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	19.5	21.7	27.4	41.9	79.5	179.5	444.8
0.0231	22.5	24.7	30.4	45.1	82.8	182.9	448.2
0.0531	29.7	31.9	37.6	52.3	90.1	190.3	455.6
0.1225	40.4	43.4	50.7	67.1	105.1	205.5	470.9
0.2823	53.5	58.3	69.5	92.5	137.6	238.6	504.3
0.6507	63.2	71.6	90.8	128.3	193.9	313.4	580.6
1.5000	65.9	79.5	110.3	170.3	274.6	447.6	754.7

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	23.1	25.4	31.4	46.1	83.1	179.6	434.5
0.0231	25.1	27.5	33.5	48.4	85.5	182.1	437.1
0.0531	31.1	33.4	39.4	54.5	91.8	188.6	443.6
0.1225	44.8	47.6	54.5	69.7	107.1	204.1	459.3
0.2823	63.9	68.1	78.5	100.9	143.6	240.4	495.7
0.6507	88.2	94.5	110.0	143.9	209.4	325.9	580.4
1.5000	118.2	127.2	149.5	199.4	298.3	477.2	778.4

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	10.3	12.8	20.2	40.3	93.5	235.2	611.0
0.0231	10.4	12.9	20.2	40.3	93.5	235.2	611.0
0.0531	11.2	13.5	20.4	40.3	93.5	235.2	611.0
0.1225	16.2	18.6	25.2	42.7	93.6	235.2	611.0
0.2823	26.6	29.7	37.6	56.4	101.3	235.5	611.0
0.6507	51.1	55.1	65.4	88.5	136.5	253.6	611.4
1.5000	106.5	110.2	121.7	151.3	212.5	336.8	648.5

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	13.7	16.4	23.5	41.8	89.0	214.1	545.3
0.0231	13.7	16.4	23.5	41.8	89.0	214.1	545.3
0.0531	13.9	16.4	23.4	41.8	89.0	214.1	545.3
0.1225	20.9	23.3	28.8	44.0	89.0	214.1	545.3
0.2823	34.9	38.6	46.8	63.4	98.9	214.2	545.3
0.6507	58.3	63.9	76.9	102.6	148.0	239.9	545.3
1.5000	97.9	106.6	126.5	166.6	237.5	356.2	599.5

A.0 nor2x4

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	18.3	19.4	22.4	29.7	48.2	97.2	226.9
0.0231	21.4	22.5	25.4	32.8	51.4	100.5	230.2
0.0531	28.5	29.6	32.6	39.9	58.6	107.8	237.6
0.1225	38.7	40.3	44.3	53.5	73.5	122.8	252.8
0.2823	51.0	53.6	60.0	73.9	101.2	155.8	286.1
0.6507	59.8	64.3	75.3	98.9	142.6	219.0	362.1
1.5000	65.4	72.6	90.0	126.9	195.8	315.2	513.2

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	20.7	21.9	24.8	32.1	49.9	95.6	215.0
0.0231	22.8	23.9	26.9	34.3	52.3	98.0	217.5
0.0531	28.8	29.8	32.7	40.2	58.3	104.3	223.9
0.1225	41.3	42.8	46.5	55.0	73.2	119.3	239.1
0.2823	57.8	60.0	65.5	78.4	104.8	155.0	274.8
0.6507	77.6	80.7	88.9	108.0	148.1	224.3	358.3
1.5000	99.3	103.8	115.4	143.0	201.8	316.9	520.3

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	9.1	10.4	13.9	23.5	49.4	118.4	301.6
0.0231	9.3	10.5	13.9	23.5	49.4	118.4	301.6
0.0531	10.2	11.3	14.4	23.7	49.4	118.4	301.6
0.1225	15.1	16.3	19.6	28.0	51.1	118.4	301.6
0.2823	25.4	26.9	31.0	41.2	64.3	123.8	301.6
0.6507	50.1	52.0	57.2	70.1	97.6	156.8	313.4
1.5000	105.3	106.8	111.7	126.7	162.7	236.3	389.1

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	11.9	13.2	16.6	25.3	47.5	105.8	260.0
0.0231	11.8	13.1	16.5	25.3	47.5	105.8	260.0
0.0531	12.3	13.5	16.7	25.2	47.5	105.8	260.0
0.1225	19.5	20.8	23.8	30.6	49.3	105.8	260.0
0.2823	32.7	34.6	39.3	49.3	68.5	113.5	260.0
0.6507	54.5	57.5	64.8	80.5	110.2	162.2	279.0
1.5000	92.2	96.7	107.8	131.9	177.9	258.9	392.7

A.0 maj3x1

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	73.8	79.0	92.4	126.0	211.2	435.8	1030.3
0.0231	75.9	81.1	94.7	128.9	214.6	439.5	1034.0
0.0531	81.4	86.6	100.3	134.9	221.4	446.8	1041.7
0.1225	95.0	100.0	113.5	148.0	235.1	461.4	1056.8
0.2823	124.5	130.1	144.3	178.4	265.3	492.4	1088.8
0.6507	162.0	170.5	191.5	238.2	334.7	561.4	1158.6
1.5000	202.6	215.4	246.8	315.4	452.5	720.8	1317.9

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	60.8	64.9	75.3	100.8	162.5	319.6	729.5
0.0231	62.3	66.4	76.9	102.5	164.3	321.5	731.4
0.0531	66.4	70.6	81.3	107.2	169.5	326.9	736.9
0.1225	78.0	82.2	93.1	119.5	182.5	340.5	750.9
0.2823	108.7	113.4	125.0	151.2	214.2	373.0	784.1
0.6507	152.9	159.6	176.2	214.0	290.4	450.1	861.6
1.5000	207.5	217.0	241.1	296.3	409.7	624.4	1043.1

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	50.8	56.7	72.8	116.5	231.1	536.0	1344.1
0.0231	50.7	56.6	72.7	116.4	231.0	535.8	1343.9
0.0531	50.6	56.5	72.7	116.3	231.0	535.8	1343.8
0.1225	50.8	56.6	72.7	116.3	230.9	535.7	1343.8
0.2823	57.7	62.6	76.7	117.5	231.0	535.7	1343.8
0.6507	82.4	88.6	105.6	148.0	244.9	535.9	1343.8
1.5000	134.0	141.2	160.7	209.9	323.9	582.6	1344.5

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	53.8	57.7	68.3	97.0	172.4	374.0	909.9
0.0231	53.8	57.6	68.3	97.0	172.4	374.0	909.9
0.0531	53.7	57.6	68.2	96.9	172.4	374.0	909.9
0.1225	52.0	56.0	67.2	96.6	172.3	374.0	909.8
0.2823	63.6	66.5	75.1	100.1	172.0	373.9	909.8
0.6507	92.4	97.7	110.7	139.6	199.8	378.1	909.7
1.5000	143.9	151.7	171.3	215.2	299.6	460.7	922.4

A.0 maj3x2

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	70.1	72.8	79.6	96.7	139.2	249.3	539.2
0.0231	72.1	74.7	81.7	99.1	142.1	252.7	542.8
0.0531	77.9	80.4	87.3	104.9	148.4	259.7	550.4
0.1225	91.9	94.4	101.1	118.3	161.9	274.0	565.5
0.2823	122.2	125.1	132.5	150.2	193.2	305.2	597.6
0.6507	161.9	166.2	177.2	203.2	259.1	376.6	668.8
1.5000	208.6	215.2	231.9	270.8	352.2	515.4	833.7

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	52.9	54.8	59.8	72.3	102.0	175.5	363.5
0.0231	54.4	56.3	61.4	73.9	103.8	177.4	365.4
0.0531	58.6	60.5	65.7	78.5	108.7	182.6	370.8
0.1225	70.5	72.4	77.5	90.5	121.1	195.7	384.4
0.2823	99.1	101.4	107.4	121.8	152.6	227.2	416.7
0.6507	137.0	140.3	148.9	169.6	214.1	303.0	492.6
1.5000	179.8	184.5	196.9	226.8	292.3	424.8	668.6

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	47.0	49.8	57.7	78.8	134.5	282.8	675.5
0.0231	46.9	49.8	57.6	78.7	134.4	282.7	675.3
0.0531	46.8	49.7	57.5	78.6	134.4	282.6	675.3
0.1225	47.0	49.8	57.5	78.6	134.4	282.6	675.2
0.2823	54.2	56.5	63.0	81.9	134.9	282.6	675.2
0.6507	77.1	80.2	88.6	110.6	161.3	291.2	675.2
1.5000	126.8	130.4	140.1	165.5	226.7	367.4	704.4

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	47.1	48.8	53.6	66.5	100.5	191.2	432.5
0.0231	47.0	48.8	53.6	66.5	100.5	191.2	432.5
0.0531	46.9	48.6	53.5	66.4	100.5	191.1	432.5
0.1225	45.5	47.2	52.1	65.3	100.2	191.1	432.5
0.2823	58.5	60.2	64.7	74.8	104.1	190.7	432.4
0.6507	86.1	88.7	95.6	111.5	145.0	216.5	434.4
1.5000	135.4	139.3	149.4	173.4	224.4	321.0	508.0

A.0 maj3x4

cell_rise (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	68.4	69.8	73.3	82.2	104.0	159.2	302.8
0.0231	70.4	71.7	75.2	84.2	106.4	162.3	306.3
0.0531	76.1	77.4	80.9	89.8	112.2	168.7	313.5
0.1225	90.2	91.4	94.7	103.5	125.5	182.2	327.8
0.2823	120.0	121.5	125.3	135.0	157.2	213.2	359.1
0.6507	158.6	160.7	166.5	180.9	213.2	282.3	430.3
1.5000	204.7	208.0	216.8	238.3	286.3	385.9	582.5

cell_fall (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	51.2	52.1	54.7	61.1	76.8	114.8	209.2
0.0231	52.6	53.6	56.1	62.7	78.4	116.5	211.0
0.0531	56.6	57.6	60.2	66.9	82.9	121.3	216.1
0.1225	68.2	69.2	71.7	78.4	94.5	133.5	228.9
0.2823	94.9	96.1	99.3	107.1	125.0	163.8	259.6
0.6507	128.7	130.3	134.8	145.9	171.6	226.4	333.5
1.5000	163.8	166.2	172.5	188.4	225.3	305.7	463.8

rise_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	45.2	46.6	50.4	60.8	88.2	161.4	355.3
0.0231	45.1	46.5	50.3	60.6	88.1	161.3	355.2
0.0531	45.0	46.4	50.2	60.6	88.0	161.2	355.1
0.1225	45.1	46.5	50.3	60.6	88.0	161.2	355.1
0.2823	53.0	54.0	57.1	65.9	90.7	161.5	355.1
0.6507	75.8	77.4	81.5	92.6	120.5	183.7	359.5
1.5000	125.9	127.7	132.5	145.3	177.6	255.2	428.1

fall_transition (ps)

$t_{in} \setminus C_L$	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.0100	46.1	46.9	49.3	55.6	72.3	116.8	235.2
0.0231	46.0	46.9	49.2	55.6	72.3	116.8	235.2
0.0531	45.9	46.7	49.1	55.5	72.2	116.7	235.2
0.1225	44.6	45.5	47.8	54.2	71.2	116.5	235.1
0.2823	58.1	58.9	61.2	67.1	80.3	119.5	234.9
0.6507	84.5	85.9	89.5	98.4	118.2	159.9	254.6
1.5000	131.6	133.6	138.8	151.8	181.7	244.0	360.5

B Python Characterization Scripts

The full automation scripts are hosted in the project GitHub repository at:

https://github.com/Sama3mad/Project-1---Standard-Cell-Library-Design-and-Characterization-.git

The repository contains the following key files:

- `cells.lib` — SPICE subcircuit definitions for all 13 cells.
- `verify.sp` — logic verification testbench.
- `characterize.py` — main characterization script for inverters, NAND2, and NOR2 cells.
- `char_maj3.py` — MAJ3 characterization script with corrected B = 1, C = 0 sensitization.
- `nldm_results.npy` — extracted NLDM data for all 13 cells in NumPy format.
- `inverter_delay_vs_load.png` — delay vs. load plot for the inverter family.

The complete listings of `characterize.py` and `char_maj3.py` are included in Section 4 of this report.